

Name: Pratik Patel
class DISC
Assignment 2

Create a Rest API with the Serverless Framework

The following steps should be followed:-

Step 1: Install serverless framework
Install the serverless framework by using npm
install -g serverless.

Step 2: Create a new service

Navigate to the directory where you want to create the project then create a new serverless service
Serverless create --template aws-nodejs --path my-rest-api

Step 3: Install dependencies

if using Node.js you may want to install additional libraries like uuid for generating unique identifiers for resources.

npm init -y

npm install uuid.

Step 4:- Define functions in serverless.yml open the serverless.yml file here we use the configuration for REST API with two routes (GET/users and POST/users).

Service: my-rest-api

provider:

name: aws

runtime: nodejs14.x

region: us-east-1

functions:-

getUsers:

handler: handler.getUsers.

events:

- http.

path: users

method: get

cors: true.

Create User:

Handler: handler.createUser

events:

- http.

path: users.

method: post

cors: true

plugins

- serverless-offline

Step 5: Create Handlers in handler.js. Define the logic for API endpoints. Create and open handler.js file

service: my-swt-api

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functions:

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handler: handler.getUsers

events:

- http:

path: users.

method: post

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Plugins:

- Serverless - Offline

Step 5: Create Handlers in handler.js Define the logic for API endpoints. Create and open handler.js file.

Step 6: Deploy to AWS.

Run the following command to deploy your API to AWS.

Step 7 Test with a postman or curl.

```
Curl -X POST https://<api-id>.execute-api:<region>.amazonaws.com/dev/users/
```

```
-H "Content-Type: application/json" \
```

```
-d '{"name": "pratik"}'
```

This will return a json response with new user.

Case study for sonarqube

Creating a custom profile in sonarqube for analysing and testing project quality involves following steps:

Step 1: Install sonarqube

- Download and install SonarQube on your local machine or server. Ensure Java is installed.

- Access SonarQube at <http://localhost:9000>.

Step 2: login and create a custom profile
login as administrator (admin/admin) as default credentials.

Step 3: Create a Custom profile you can copy an existing profile (eg. sonarway profile) as a base or start from scratch.

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Step 3: Assign Custom profile to Project.

- Go to projects, select the project you want to analyze and assign your newly created quality profile to it.

Step 4: Analyze Project with SonarQube

Run the SonarQube scanner on your projects.

Using Sonarcloud to analyze Github code

Step 1: Setup a Sonarcloud account

- Go to Sonarcloud and sign up using your Github account.

Step 2 Import Github rep.

Step 3: Configure Sonarcloud analysis.

Step 4 Add Github Action for continuous analysis.

Step 5 Analyze Results.

Analyze Python project with SonarQube

Step 1 Install SonarQube Python Plugin

Step 2 Configure Sonar-project properties.

Step 3 Run SonarQube Scanner

Step 4 Review Python Analysis.

Analyze Node.js Project with SonarQube.

Step 1: Install Node.js plugin

Step 2: Configure Sonar project properties.

Step 3 Run SonarQube Scanner

Step 4: Analyze Results.

Terraform and self serve infrastructure model for large organizations.

Using Terraform for self serve infrastructure in large organization the operations team often get repetitive infrastructure request. A solution to streamline this is by using terraform to build a self aware infrastructure model with Terraform. product teams can independently manage their infrastructure with original standard

Benefits

Allows decentralized teams to deploy services efficiently.

Ensures compliance with best practices and standard through reusable terraform modules.

Reduces the workload on the central operations.

~~Terraform modules : By creating reusable Terraform modules, you can encapsulate best practices for deploying services like database, apps or containers.~~

Terraform cloud and Ticketing system integration. terraform cloud integrates with ticketing system like service now to automate infrastructure requests.